

MES + SCADA + Virtual Factory — Source of Truth

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Architecture, phases, industrial integration, and MES design.

Revision 2026-08-24. Living architecture: change this document *intentionally*, then regenerate the PDF.

How to update this PDF: edit docs/source/MES_SCADA_Virtual_Factory_Source_of_Truth.md, then run docs/source/generate-sot-pdf.sh. Do not maintain a second competing Source of Truth.

Documentation hierarchy: SoT PDF → implementation-status.md → architecture.md → decisions.md → roadmap.md → CHANGELOG.md → docs/README.md.

Status labels used throughout: IMPLEMENTED · PARTIALLY IMPLEMENTED · VALIDATED · PLANNED · NOT IMPLEMENTED. Planned text is architecture, not code.

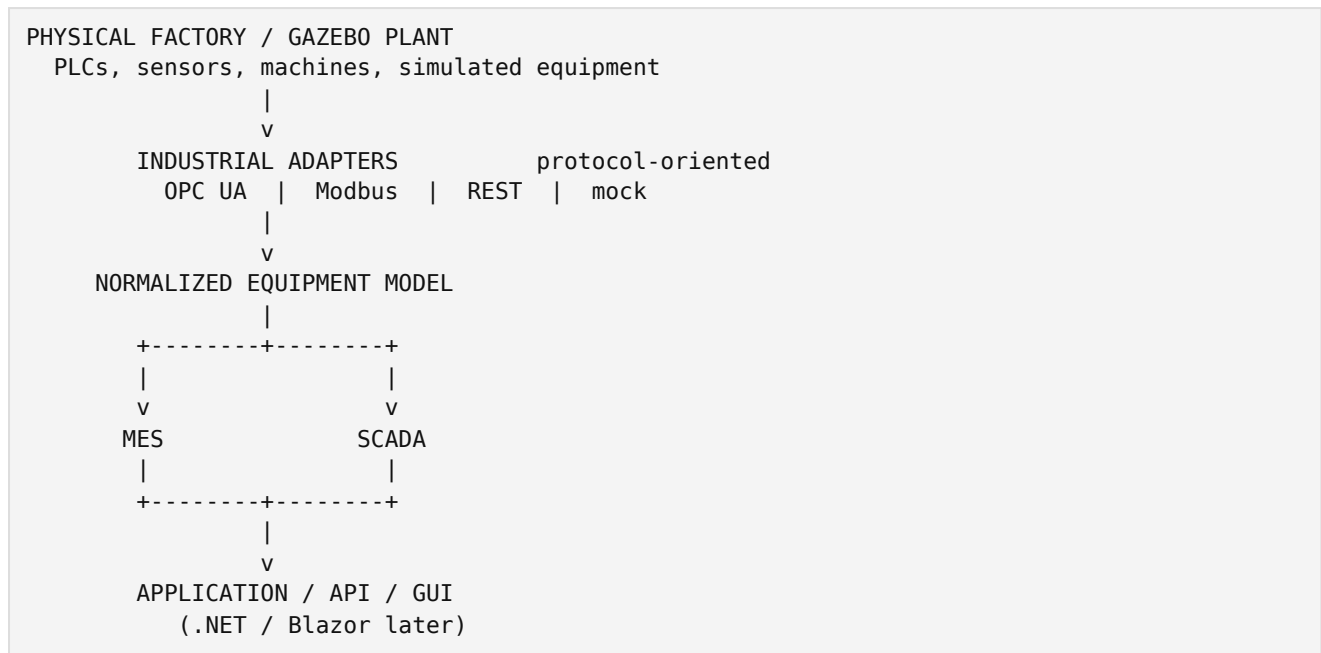
1. Purpose

This document is the project's **only active architectural Source of Truth**. It preserves terminology, phase structure, industrial-integration rules, and the intended MES so later work can resume without inventing a second architecture.

Gazebo is a **simulation plant**. C++ plugins are **not** the GUI. Industrial adapters are **not** the MES. SCADA is **not** MES.

When implementation and this document disagree: **current source code** is implementation reality; **this PDF** is architectural authority. Resolve the conflict with an ADR and an intentional SoT update. Do not silently drift.

2. Executive architecture



The Industrial Adapter is the boundary between heterogeneous factory equipment and the application. MES must not understand every PLC vendor, NodeId, Modbus map, or Gazebo ECM.

3. What we are building

Layer	Role	Status
Virtual factory	Gazebo representation of a plant	IMPLEMENTED (Phases 1-4)
Equipment control	C++ Gazebo System plugins	IMPLEMENTED (conveyor example)
Equipment abstraction	Open-ended technical asset model	IMPLEMENTED (Phase 5)
Industrial adapter layer	Protocol connectors into Equipment	PARTIALLY IMPLEMENTED (Phase 6: mock + OPC UA; Modbus/REST NOT IMPLEMENTED)
MES	Production execution and manufacturing information	NOT IMPLEMENTED (Phase 7)
SCADA / HMI	Live monitoring, alarms, operator control	NOT IMPLEMENTED (Phase 8)
Security	Authentication, RBAC, audit	NOT IMPLEMENTED (Phase 9)
Real factory	Production devices through adapters	NOT IMPLEMENTED (Phase 10)
Commercial / enterprise	Hardening, ERP/PLM/QMS/CMMS, advanced intelligence	NOT IMPLEMENTED (Phase 11)

4. Official phases (1-11)

This numbering is authoritative. Do not revive Stage 0–25, sensor-first, or gateway-only plans.

Phase	Name	Status
1	Factory Foundation	IMPLEMENTED
2	Equipment Plugin Foundation	IMPLEMENTED
3	Conveyor Control	IMPLEMENTED
4	Product Motion	IMPLEMENTED
5	Industrial Equipment Abstraction	IMPLEMENTED
6	Industrial Adapter Layer	IN PROGRESS — architecture + mock + OPC UA IMPLEMENTED ; Modbus/REST NOT IMPLEMENTED
7	MES Core + Resource Management	PLANNED / NOT IMPLEMENTED
8	SCADA / Operational HMI	PLANNED / NOT IMPLEMENTED
9	Security & Authorization	PLANNED / NOT IMPLEMENTED
10	Real Factory Integration	PLANNED / NOT IMPLEMENTED
11	Commercial Hardening & Enterprise Integration	PLANNED / NOT IMPLEMENTED

Capability mapping (all **PLANNED** unless noted):

- Dynamic plant configuration, PLC/equipment onboarding, hierarchy, work centers, resource management, orders, materials, scrap, quality, genealogy, OEE, downtime, scheduling, personnel, tools, events, operational analytics → **Phase 7**
- Live visualization, shop-floor execution views, alarm HMI → **Phase 8**
- Identity, authentication, RBAC, organizational permissions → **Phase 9**
- Production-grade connection of real PLCs/machines (no MES rewrite) → **Phase 10**
- ERP/PLM/QMS/CMMS, advanced what-if/optimization, cost accounting, deployment/observability → **Phase 11**

Do not implement Phase 7 until explicitly instructed. Remaining Phase 6 work (Modbus, REST) is still open.

5. Current implementation (facts)

IMPLEMENTED

- Gazebo plant: world, floor, CV-001, PRODUCT-001, conveyor plugin, kinematic product motion (dt in seconds).
- Equipment / GenericEquipment / Conveyor (Gazebo example only).

- IndustrialAdapter / MockIndustrialAdapter / OpcUaIndustrialAdapter (open62541 1.4.0-rc2).
- One OPC UA adapter instance = one UA_Client = one endpoint (ADR-026).
- Multi-server composition of adapter instances.
- OPC UA multi-server **validation** at 10, 25, 50, 100, and 200 **simulated in-process** servers. **VALIDATED** under those test conditions. **Not** production hardware proof. **Not** “unlimited PLCs.”

NOT IMPLEMENTED

- MES, resource management, scheduler, OEE engine, materials, scrap, quality, genealogy, analytics engine
- Dynamic PLC management UI/API
- SCADA, GUI, authentication/RBAC, database, application API
- Modbus adapter, REST adapter
- Machine-specific classes such as Robot.hh, Pump.hh, Oven.hh

Development Start/Stop at simulation updates 1000/5000 is a **temporary test**, not industrial control.

6. Equipment model (Phase 5) — IMPLEMENTED

Equipment is an **open-ended technical asset**, not a catalog of C++ machine classes.

A plant may contain conveyors, robots, pumps, motors, CNC, welding cells, ovens, furnaces, mixers, inspection, packaging, manual stations, test equipment, AGVs/AMRs, storage, custom machines, and **unknown future types**. Represent ordinary machines as GenericEquipment with identity, type metadata, capabilities, commands, telemetry, state, and fault. Add a specialized C++ class only when behaviour cannot be expressed that way (Conveyor is the Gazebo exception).

MES/SCADA must not switch on type().

Equipment concepts: identity, type metadata, operational state (Stopped/Running), commands (execute), telemetry {name, value, unit}, machine fault, adapter connection state **as a separate comms signal**.

Do **not** put scheduling, allocation, reservation, capacity, or plant hierarchy on Equipment.

7. Industrial adapters (Phase 6) — PARTIALLY IMPLEMENTED

Adapters are **protocol-oriented**:

- OpcUaIndustrialAdapter — **IMPLEMENTED**
- MockIndustrialAdapter — **IMPLEMENTED**
- ModbusIndustrialAdapter — **NOT IMPLEMENTED**
- RestIndustrialAdapter — **NOT IMPLEMENTED**

Not: PumpPLCAdapter, RobotPLCAdapter, ConveyorPLCAdapter, SiemensAdapter.

```

one OpcUaIndustrialAdapter instance
|
+-- one UA_Client
|
+-- one OPC UA server / endpoint
|
+-- mapped GenericEquipment

```

N OPC UA PLCs ⇒ N adapter instances. Adding a PLC must **not** require a new C++ adapter class. Configuration creates another instance and mappings.

OPC UA mapping is C++ config today (OpcUaAdapterConfig). Future MES onboarding (Phase 7) stores equivalent configuration and instantiates adapters **without recompiling** application code.

REST remains a fallback for gateways/vendor APIs. It does not replace OPC UA or Modbus.

Test OPC UA security (SecurityPolicy#None, anonymous localhost) is **DEVELOPMENT ONLY**.

8. MES vs Equipment vs Adapter

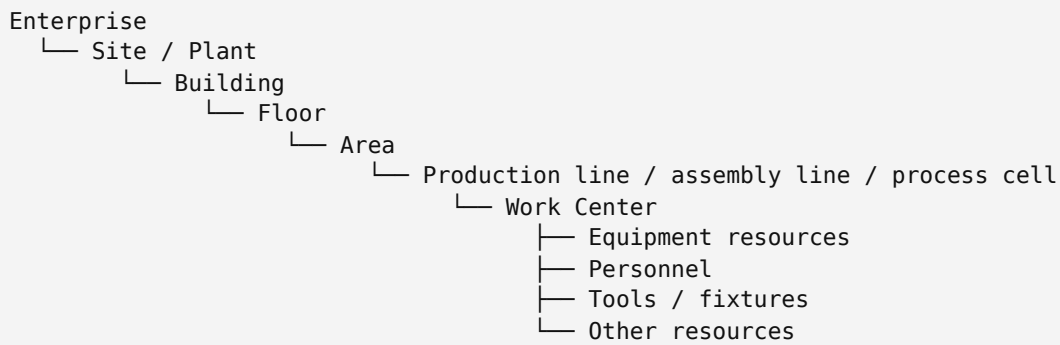
Concern	Owner	Status
Technical asset, telemetry, commands, machine fault	Equipment	IMPLEMENTED
Protocol, endpoint, comms lifecycle	IndustrialAdapter	PARTIALLY IMPLEMENTED
Capability, availability, allocation, reservation, capacity, readiness	MES Resource Management	PLANNED
Orders, materials, quality, genealogy, OEE, schedule	MES	PLANNED
Live HMI, alarms, operator commands	SCADA	PLANNED
Login, RBAC	Phase 9	PLANNED

A machine may be technically Running and still **MES-unavailable** (reserved, maintenance window, quality hold). Do not confuse Equipment::fault() with MES availability.

9. Configurable plant hierarchy — PLANNED (Phase 7)

Organizational structure is **data**, not C++ inheritance.

Minimum conceptual tree (all node types optional/configurable):



Valid alternatives include Plant → Area → Process Cell → Work Center, or Plant → Building → Floor → Line → Work Center. The MES stores parent/child location entities and assignments. “Assembly line” is not the only allowed production structure.

Work Center is a **production capability**, not necessarily one machine. Example: WC-100 may combine Robot-17, Welder-03, Operator-22, Fixture-4, with constraints (qualification, fixture, maintenance, material, capacity).

10. Dynamic PLC / equipment onboarding — PLANNED (Phase 7)

Real scenario: a plant has 100 PLCs; six months later a new line adds 100 more. The administrator must **not** modify and recompile C++ merely to add those PLCs.

Future configuration/UI/API (Phase 7, UI later with Phase 8/9) must allow:

- Add a PLC / industrial source (identity, protocol, endpoint, connection parameters)
- Define equipment represented by that source and node/register mappings
- Assign plant / building / floor / area / line / work center
- Assign responsible supervisor/manager
- Assign capabilities and related resources
- Enable/disable connection, test connectivity, monitor health

For OPC UA:

```
PLC / OPC UA server
  v
OpcUaIndustrialAdapter instance (created from config)
  v
Normalized Equipment
  v
MES (resource + location assignment)
```

No new C++ adapter class per PLC. Phase 9 later restricts who may change configuration.

11. Supervisors and responsibility — PLANNED

The MES data model must record responsibility/ownership, for example Plant Manager, Area Manager, Production Manager, Line Supervisor, Work Center Supervisor, Maintenance Supervisor, Quality Supervisor. A person may be responsible for multiple resources.

Authentication/RBAC is Phase 9 and is not implemented. Operational qualification (may this operator weld?) is a Phase 7 resource concept and is distinct from login permission.

12. MES resource availability — PLANNED

Equipment technical state (examples): Stopped, Running, Faulted.

MES availability (examples): Available, Unavailable, Reserved, Allocated, Maintenance, Blocked, Quality Hold, Scheduled, Decommissioned.

These are different axes.

13. Resource readiness before dispatch — PLANNED

Before dispatch, MES evaluates required resources. Result is **READY** or **HOLD** with **specific reasons**, not a generic “resource unavailable.”

Example holds: equipment unavailable; equipment faulted; maintenance active; operator unavailable; qualification expired; material shortage; material quality hold; tool unavailable; work-center capacity exhausted; line committed; incompatible setup; resource reserved by another order.

14. Production orders — PLANNED (Phase 7)

Orders, work orders, operations, routing, BOM, bill of process, dependencies, required resources/capabilities/materials/personnel/tools, quantities, planned/actual times, priority, due date, status, holds, completion, partial completion, scrap, rework.

Do not implement in this documentation pass.

15. Materials — PLANNED (Phase 7)

First-class MES capability: raw materials, components, subassemblies, WIP, finished goods, consumables, packaging.

Track identity, lot/batch, serial where applicable, quantity/unit, location, status, quality status, expiry, reservations, consumption, production, scrap, return, transfer.

Material availability participates in scheduling and readiness.

16. Scrap and waste — PLANNED (Phase 7)

Track planned / good / scrap / rework quantities; scrap reason and category; operation, work center, equipment, material lot, operator, timestamp, order, product.

Do **not** compute factory efficiency from machine runtime alone. Scrap and material waste must be visible.

17. OEE — PLANNED (Phase 7)

OEE = Availability × Performance × Quality.

- Availability: planned production time vs downtime (planned, unplanned, maintenance, faults, configured changeover).
- Performance: ideal vs actual cycle time, counts, speed losses.
- Quality: total / good / rejected / scrap / rework.

Provide OEE views at Equipment, Work Center, Line, Area, and Plant **using defined aggregation rules**. Do **not** blindly average OEE percentages across machines. Prefer count-weighted or time-weighted rollups of the underlying components, documented per KPI.

Keep OEE mathematically distinct from broader plant efficiency (yield, material variance, labor utilization, on-time completion).

18. Downtime — PLANNED (Phase 7)

Downtime events with start/end/duration, resource, reason tree (mechanical, electrical, controls, material, operator, quality, changeover, setup, maintenance, waiting, starvation, blockage, unknown), category, order/operation, operator, source, acknowledgement, comments.

19. Quality — PLANNED (Phase 7)

Inspection plans, characteristics, specifications, sampling, measurements, pass/fail, defects, nonconformance, rework, scrap, corrective actions, quality holds, release. SPC (control charts, limits, trends, out-of-control detection) is a **planned** capability, not Phase 7 code today.

20. Genealogy / traceability — PLANNED (Phase 7)

Forward and backward genealogy: finished product → consumed lots, operators, equipment, work center, inspections; and material lot → products that consumed it. Required for recall and root-cause analysis.

21. Events and analytics layers — PLANNED

Distinguish:

1. **Raw industrial data** (OPC UA nodes, later Modbus/REST)
2. **Contextualized MES facts** (order, operation, work center, material, person)
3. **Aggregated analytics** (hour/day/week/month/year KPIs)

Dashboards must not compute everything from raw tags.

Preserve historical events such as EquipmentStateChanged, DowntimeStarted/Ended, ProductionStarted/Completed, MaterialConsumed/Produced, ScrapRecorded, InspectionCompleted, NonconformanceCreated, ResourceAllocated/Released, OrderDispatched/Completed — attributable to time, plant, line, work center, equipment, order, operation, material, person where applicable.

Analytics periods: real-time, hourly, daily, weekly, monthly, yearly. Each period compares to target, previous period, plan, and historical baseline.

Example KPIs (all **PLANNED**): Material Yield, Scrap Rate, First Pass Yield, Rework Rate, Material Variance, Production Yield, Resource Utilization, Labor Utilization, Capacity Utilization, OEE, Throughput, On-Time Completion.

Loss analysis (downtime, speed, scrap, rework, material, labor, capacity, changeover) may later estimate cost via ERP. Phase 7 does **not** require a finance engine.

22. Scheduling — PLANNED (Phase 7)

Planning, scheduling, dispatching, rescheduling using priority, due date, resource/equipment/work-center capability, material, operator, qualification, tooling, maintenance windows, changeover, sequence-dependent constraints, capacity, operation dependencies.

The scheduler should explain why a schedule was chosen or why an order cannot be scheduled. Advanced what-if / optimization may extend into Phase 11.

Bottleneck analysis: which work center/equipment/material/labor/tool is constraining, why, duration, affected orders.

23. Maintenance, personnel, tools — PLANNED

MES observes maintenance availability (planned windows, unplanned, status, related downtime). A full CMMS is **not** Phase 7; CMMS integration is Phase 11.

Personnel: roles, qualifications, certifications, shifts, availability, assignment, labor utilization. Distinct from Phase 9 RBAC.

Tools/fixtures: availability, assignment, reservation, compatibility, usage, calibration/maintenance status.

24. SCADA (Phase 8) — PLANNED

Live state, commands, alarms/ack, trends, role-specific operational screens. Consumes Equipment + MES context. Not a substitute for MES. Not a C++ desktop inside the Gazebo plugin.

25. Security (Phase 9) — PLANNED

Authentication, RBAC, audit, least privilege. Supervisor is the floor-management role name. Do not scatter permission if-statements in equipment/adapters.

26. Real factory (Phase 10) — PLANNED

Connect real PLCs/sensors/machines through production adapters without rewriting MES. Uses the same one-adapter-per-source rule.

27. Commercial hardening & enterprise (Phase 11) — PLANNED

Configuration, deployment, observability, testing, backups, support. ERP/PLM/QMS/CMMS integration where appropriate. Advanced manufacturing intelligence beyond core MES analytics.

28. Siemens Opcenter — capability benchmark only

Siemens Opcenter is an **external industry benchmark** for production execution, tracking, resource utilization, qualifications, materials, work orders, scheduling, constraints, visibility, OEE, downtime, quality, NCR, root-cause, genealogy, SPC, KPIs, closed-loop feedback, and enterprise integration.

This project does **not** claim Opcenter feature parity and does **not** copy proprietary architecture. Those capabilities are **PLANNED** industry-aligned goals, not implemented software.

29. GUI and application stack — PLANNED

C++ remains for Gazebo plugins and industrial/performance code. Application GUI is web/.NET; Blazor is the documented candidate. Do not build the product MES/SCADA GUI as a large C++ desktop.

30. Simulation notes (IMPLEMENTED)

CV-001 is static; PRODUCT-001 is dynamic. Product motion is pose integration, not a physically accurate belt. Geometry and the 1000/5000 development timers remain as previously decided until deliberately changed.

Gazebo plugin is **not** an `IndustrialAdapter`.

31. Decisions to preserve

1. Gazebo is the virtual plant, not MES/SCADA.
2. C++ is for simulation and industrial/performance code, not the main MES GUI.

3. Adapters isolate MES/SCADA from vendors and protocols.
 4. Multiple protocols; REST is a fallback, not a replacement.
 5. MES consumes normalized Equipment and MES resources, not Nodelds or Gazebo ECM.
 6. One OPC UA adapter instance per OPC UA server.
 7. Equipment $\neq$ MES resource.
 8. Plant hierarchy is configurable data.
 9. Adding PLCs is configuration, not a new C++ class per PLC.
 10. Resource readiness reports specific constraint reasons.
 11. Materials and scrap participate in readiness and analytics.
 12. OEE is distinct from broader efficiency; do not blindly average OEE %.
 13. Analytics use contextualized events, not raw tags alone.
 14. Genealogy is forward and backward.
 15. Opcenter is a benchmark, not a claim of parity.
 16. There is one SoT. Update it on purpose.
 17. Development timers are not industrial control.
 18. Security includes authentication, authorization, and audit (Phase 9).
 19. Supervisor is the floor-management role name.
 20. Do not implement Phase 7 until instructed.
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32. Next direction

Now: Phase 6 remaining protocol adapters (Modbus, REST) when instructed; **do not start MES code** in this increment.

Then: Phase 7 MES Core + Resource Management as specified above, still consuming the existing Equipment and adapter contracts.

If a future choice conflicts with this document, revise the Source of Truth deliberately and regenerate this PDF.